# Renuka SINGH VIRK

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## Profile

Swiss M.Sc. student in Electrical & Electronic Engineering at EPFL with hands-on experience in AI, machine learning, optimization and signal processing.
I'm motivated by technically challenging work with societal relevance, and I value the intersection of engineering, ethics, and broader human impact.

## Education

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| 2023 - now | MSc in Electrical and Electronic Engineering at **EPFL** |
| 2024 | 21-ECTS bridging program to transition into the Master's in Electrical Engineering at **EPFL** |
| 2020 - 2023 | BSc in Life Sciences Engineering at **EPFL** |
| 2018 - 2019 | High School academic exchange at **Kantonsschule Romanshorn** (Thurgau, Switzerland) |
| 2008 - 2020 | Classical piano and music theory certificate at **Conservatoire de Lausanne** |

## Professional experience

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| 2026 | AI & Data Science Intern at **Logitech Europe SA**; developing agent-based systems to automate data analysis workflows and support decision-making |
| 2021 - now | First Aid Instructor at **FirstMed SA**, Basic Life Support (BLS) and Pediatric BLS proficiency |
| 2023 - 2024 | Teaching Assistant for the courses of Electromagnetism and General Mathematics II at **EPFL** |
| 2025 | Teaching Assistant for the course of Electrical Systems and Electronics I (EE295) at **EPFL** |
| 2021 - 2023 | Mentor at **EPFL**, charged of helping a group of 1st-year BSc students accommodate to university |

## Projects

**Ask the Migration Reports: A RAG Chatbot (Python)** IDIAP

Developed a RAG chatbot in the context of the EU Commission ELIAS Project (Use Case 4). The chatbot helps support migrants and refugees by facilitating access to official migration reports (EMN Country Factsheets) and provides answers with sources to the data used.

**AI-Driven Transformation of Hate-Speech Video Content (Python)** EE-559

Deployed a multimodal deep-learning model (BERT + Meta-Llama-3-8B) to detect and rephrase hateful speech in video streams.

**Data Visualization of the World Values Survey Dataset (JavaScript)** COM-480

This data visualization project takes the form of a quiz to determine which country aligns the most with an individual's values. It offers extensive visualization of the World Values Survey Dataset.

**Signal and Image Processing (Python)** EE-490(f)

Small projects introducing the most common filtering methods for image data, common edge and object detection methods including histogram of gradients, template matching, as well as tracking methods applicable to video data.

**Coin-Image Classification with AlexNet (Python)** EE-451

Designed a pipeline combining morphological segmentation and AlexNet-based classification to recognize coin denominations.

**Intrusive Memory Classification from 654 Features (Python)** CS-433 — UZH

Developed gradient-boosted models and logistic regression to distinguish PTSD vs. CUD patients based on survey data.

**Cardiovascular Risk Prediction on 328k+ Patients (Python)** CS-433

Developed and tuned logistic regression models to predict CVD risk in a highly imbalanced dataset.

**Optimal Strategy for a 6×6 Repeated Coin-Hiding Game (Python)** MGT-483

Formulated a zero-sum LP, proved König's theorem, and implemented a simplex solver delivering strategies that cap losses at 25 coins.

**OFDM Transmitter/Receiver for Acoustic Data (Matlab)** EE-442

Implemented an OFDM system for wireless acoustic transmission using BPSK/QPSK, cyclic prefixes, channel equalization, and continuous phase tracking; evaluated spectral efficiency and transmitted image data under real-world conditions.

**Mini-Games on ATmega128 (Assembly)** MICRO-210

Coded interactive mini-games in AVR assembly using keypad, LCD, temperature sensor, EEPROM, and buzzer; implemented real-time input handling, phase-based game logic, and secure passcode verification.

**Temple-Run-Style Game for Nintendo DS (C)** EE-310

Developed a 2D side-scroller in C using Nintendo DS GPU sprites, dual ARM processors, timers, touch input, and sound; implemented custom level selection, animations, and real-time collision handling.

**Ant-Colony Behaviour Simulation (C++)** CS-112

Object oriented programming model of pheromone signaling and foraging in a toroidal world.

## Skills

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| Programming | Python, C, C++, Julia, MATLAB, JavaScript |
| Hardware/RTL | Vivado (Design Suite), Assembly (AVR), VHDL |
| Languages | French (native), English (native), German (B2) |

## Personal information

**Hobbies** Travel, piano, running, drawing, hiking, jewelry making